

CityMind — Final Project Report (Phase 2)

Course: AI Project **Group Members:**

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1. Executive Summary

CityMind is an interactive Urban Intelligence System that models a mid-sized city as a weighted, undirected graph and runs five cooperating AI modules on top of a single shared CityGraph and an EventBus publish/subscribe layer. The system is packaged as a desktop application and executes a 20-step simulation in which modules react to real events (road blockages, crime-risk updates, ambulance repositioning, re-routing, rescues).

All five challenges defined in Phase 1 are implemented and running. The deliverable also includes a sixth auxiliary module (police deployment) that the risk pipeline feeds into, a full PyQt6 command-center UI, a zoom/fit control for the map, and a structured event log that can be saved to disk.

2. What We Built

2.1 Shared infrastructure

File	Purpose
core/city_graph.py	Single CityGraph object (NetworkX adjacency model). Stores node properties (type, population density, predicted risk, grid position) and edge properties (base cost, risk multiplier, is_blocked, effective_cost).
core/event_bus.py	Lightweight pub/sub (ROAD_BLOCKED, ROAD_UNBLOCKED, RISK_UPDATED, AMBULANCE_MOVED, POLICE_MOVED, CIVILIAN_RESCUED, SIMULATION_STEP, REROUTE). Every module subscribes rather than polling.
simulation/runner.py	Orchestrates initialize() → 20-step

	loop → complete. Aggregates stats and exposes a step-driven API consumed by the UI.
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2.2 Challenge modules (in challenges/)

- **C1 — City Layout Planning (`c1\_layout.py`)**

CSP-style layout with a *constructive seed* (typed placements with adjacency rules), a *backtracking repair* pass and a *Min-Conflicts local-search* fallback. Tracks hard (Industrial-Adjacency, PowerPlant-Industrial) and soft (Residential→Hospital bridge) violations, enforces connectivity.

- **C2 — Road Network Optimisation (`c2\_roads.py`)**

Genetic Algorithm over candidate edge subsets. Population is **seeded from a Kruskal MST** plus forced redundant edges to satisfy the Hospital↔AmbulanceDepot dual-path constraint. Fitness = weighted sum of (total cost + connectivity penalty + redundancy penalty). 120 generations with tournament selection, elitism, crossover and mutation.

- **C3 — Ambulance Placement (`c3\_ambulance.py`)**

Simulated Annealing over placements of 3 ambulances. Energy = worst-case shortest-path distance from any citizen to its nearest ambulance, computed on a pre-cached distance matrix. Subscribes to ROAD_BLOCKED / RISK_UPDATED and re-searches automatically.

- **C4 — Emergency Routing (`c4\_routing.py`)**

A with *Euclidean heuristic + greedy nearest-unvisited civilian selection*. Listens on `ROAD\_BLOCKED`; triggers an instant re-plan* from the team's current node to the next target and publishes a REROUTE event.

- **C5 — Crime Risk Prediction (`c5\_crime.py`)**

Feature engineering (population density, industrial proximity, location type). **K-Means** with elbow inertia logging and **Decision Tree** classifier trained on synthetic labels. Resulting risk class (High / Medium / Low) is written back to nodes and **translated into edge cost multipliers** (1.5× / 1.2× / 1.0×) picked up by C3 and C4.

- **Police Deployment (`c\_police.py`)** — auxiliary

Subscribes to RISK_UPDATED and distributes 10 officers across risk zones proportional to risk score, publishing POLICE_MOVED.

2.3 User Interface (ui/display.py)

- PyQt6 single-window command-center interface with three panels (canvas / controls / event-log) driven by a QSplitter.
- **Canvas** (GridCanvas): custom QPainter scene with per-type detailed tiles (hospital, school, industrial chimneys with animated smoke, power-plant heat shimmer, residential houses, ambulance depot radar sweep), layered road rendering (shoulder → asphalt → edge highlight → yellow dashed centerline), animated A* path, police officers, SOS civilian markers, and smooth-interpolated ambulance sprites with trails.
- **Overlays** (toggleable): Road Network, Ambulance Coverage (sonar ring), Crime Heatmap, Police Zones, A* Path, Population Density.
- **Simulation controls**: Play / Step / Reset / Re-Init, Max Steps (5–100), Speed (1–20).
- **View controls**: Zoom slider (50–300 %) and FIT WHOLE MAP button (also Ctrl + mouse-wheel) for users on any display size.
- **Stats panel**: step counter, civilians rescued, roads blocked, risk histogram, police/ambulance counts — all updated live.
- **Event Log**: colour-coded scrolling QListWidget, exportable to a text file via Save log to file.

2.4 Verification

smoketest.py boots the runner headless and walks steps. All five challenge print their phase-banner diagnostics on initialisation. Final summary emitted at simulation completion:

```
=== CITYMIND SIMULATION COMPLETE ===  
Total steps, Civilians rescued, Roads blocked/unblocked,  
Ambulance repositions, Police repositions,  
Crime refreshes, A* reroutes, Risk distribution
```

3. Deviations from Phase 1 (with justifications)

#	Phase 1 plan	Phase 2 implementation	Justification
1	UI in Tkinter or PyGame	UI in PyQt6	PyQt6 gave us proper QGraphicsView scene/viewport handling, stylesheet theming, a real signal/slot thread model for the InitThread, and high-quality antialiased painting that Tkinter

			canvas could not match. The functional contract (grid canvas, overlays, control panel, event log) stayed identical to the Phase 1 wireframe.
2	Hospital coverage = exactly "reachable within 3 road hops" (hard)	Implemented as a soft constraint (soft(Res-Hosp-bridge)) tracked in the violation counter	In C2 the GA builds roads <i>after</i> the CSP layout is fixed, so a strict 3-hop rule can make the CSP infeasible before the road graph even exists. We model the distance rule as a soft score the CSP <i>minimises</i> , which aligns with the Phase 1 "find the layout that violates the fewest rules" instruction. Hard Industrial-Adjacency and PowerPlant-Industrial rules remain strictly enforced (violations printed every run are 0).
3	Scope: C1–C5 only	Added <code>`c_police.py`</code> (police deployment)	The Phase 1 spec explicitly mentions "10 police officers" being deployed wisely; we extracted this from C5 into its own small module so the responsibility split stays clean (C5 predicts risk → police module consumes the risk map via the bus). Zero-cost to the other challenges.
4	C3 SA pure hill-climbing variant listed as an option	We use Simulated Annealing exclusively (cooling schedule: geometric, $T_0 = 1.0$, $\alpha = 0.95$)	SA empirically escapes the local optima that hill-climbing gets stuck in on the denser graphs we observed. Phase 1 explicitly listed SA as the preferred variant; we simply did not implement plain hill-climbing as a fallback because it was unnecessary.

No algorithmic choice from Phase 1 was abandoned. Everything Phase 1 promised (CSP, GA, SA, A*, K-Means + Decision Tree, shared graph, event bus, overlays, event log) is present and running.

4. Work Breakdown (2 Members)

We kept the Phase 1 split (Member 1 = constraints/planning/integration; Member 2 = search/ML/UI) and divided the new Phase 2 work along the same axis.

Member 1 — Taha Jabbar(24i-0816)

Focus: **Constraints, planning, core infrastructure, integration.**

Area	Files	Contribution
Shared graph	core/city_graph.py	Designed the CityGraph class (nodes, edges, effective-cost API, block_edge / unblock_edge publishing to the bus).
Event bus	core/event_bus.py	Implemented pub/sub, event constants, wiring patterns.
C1 — City Layout Planning	challenges/c1_layout.py	Constructive seeding, neighbour tables, Industrial-Adjacency / PowerPlant-Industrial hard rules, BFS hospital-coverage, backtrack repair, Min-Conflicts local search, populate_graph, violation reporting.
C2 — Road Network GA	challenges/c2_roads.py	Chromosome encoding, MST-seeded initial population, dual-path redundancy penalty, tournament selection / crossover / mutation / elitism, fitness logging every 20 generations.
Simulation orchestrator	simulation/runner.py	initialize() pipeline (C1 → C2 → C5 → C3 → Police → C4), 20-step loop, event counters, final summary.
System integration	across modules	End-to-end wiring: risk updates flowing into edge costs, reroute events flowing into the UI log, police deployment consuming the risk map.

Verification	smoketest.py, citymind_log.txt	Headless boot test + captured simulation log for the submission bundle.
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Member 2 — Abdullah Khalid

Focus: **Search, optimisation, machine learning, user interface.**

Area	Files	Contribution
C3 — Ambulance SA	challenges/c3_ambulance.py	Distance-matrix caching, neighbour-move operator, cooling schedule, energy function, event subscriptions for automatic re-search.
C4 — A* Routing	challenges/c4_routing.py	A* with Euclidean heuristic, greedy nearest-unvisited civilian selection, dynamic re- plan on ROAD_BLOCKED, arrival detection and CIVILIAN_RESCUED publishing.
C5 — Crime Risk	challenges/c5_crime.py	Feature engineering (density, industrial proximity, type), K- Means with elbow inertia logging, synthetic-label generation, Decision Tree training, risk → edge-multiplier integration.
Police module	challenges/c_police.py	Risk-weighted deployment of 10 officers driven by RISK_UPDATED.
UI framework	ui/display.py	Full PyQt6 command-center: theme/stylesheets, canvas painter, all per-type tile drawers, ambulance/civilian/police/team sprites, animated A* path, overlays, control panel, event- log list, save-log dialog.
View controls	ui/display.py (GridCanvas, _on_zoom, _on_fit)	Auto-fit on resize, zoom slider (50–300 %), FIT WHOLE MAP button, Ctrl-wheel zoom.
Font/robustness	main.py	_ensure_qt_font_dir, _install_windows_fonts,

		substitution map so labels render crisply across different PyQt6 installs.
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Shared responsibilities

- **Testing** — both members ran `main.py`
- **This report & the Phase 1 → Phase 2 diff table** — written jointly.

5. How to Run

```
cd "C:\Users\Fahad Computer\Desktop\CITY\citymind"
pip install -r requirements.txt # PyQt6, networkx, numpy, scikit-learn
python main.py                 # full GUI
python smoketest.py            # headless smoke test
```

In the GUI:

- **PLAY / STEP** — advance the 20-step simulation.
- **RESET** — reset the sim; **RE-INIT** — re-run C1+C2+C5+C3+Police from scratch with a new random seed.
- **Zoom slider / FIT WHOLE MAP / Ctrl + wheel** — map sizing.
- **Overlays** — toggle Road Network, Ambulance Coverage, Crime Heatmap, Police Zones, A* Path, Population Density.
- **Save log to file** — export the colour-coded event log.

6. Results Snapshot (example run, 20 steps)

```
[C1] Final layout: Hospital=3, Industrial=5, PowerPlant=3, School=4,
                  AmbulanceDepot=2, Residential=55, Empty=28
[C1] Violations: hard(Ind-Adj)=0 hard(PP-Ind)=0 soft(Res-Hosp-bridge)=0
[C2] GA 120 generations – final fitness ≈ 96-99 (100 roads built)
[C5] K-Means elbow inertias + Decision Tree accuracy ≈ 0.45-0.65
      Risk distribution: H≈14 M≈35 L≈20
[C3] Worst-case citizen distance ≈ 8-12 (grid units)
[C4] 7 reroutes triggered by 7 ROAD_BLOCKED events over 20 steps
```

All Phase 1 success criteria (valid layout, dual-path road network, fair ambulance coverage, dynamic re-planning, integrated risk multipliers) are met.

End of Report.